US 20250278057A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0278057 A1
(43) Pub. Date: Sep. 4, 2025
NOHARA

(54) IMAGE FORMING APPARATUS

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(21) Appl. No.: 19/052,230

(22) Filed: Feb. 12, 2025

(30) Foreign Application Priority Data

Feb. 29, 2024 (JP) 2024-029427

Publication Classification

(51) Int. Cl.
G03G 21/18 (2006.01)
G03G 15/00 (2006.01)
G03G 15/08 (2006.01)

(52) U.S. Cl.
CPC ***G03G 21/1853*** (2013.01); ***G03G 15/556*** (2013.01); *G03G 15/0872* (2013.01); *G03G 2215/0665* (2013.01)

(57) ABSTRACT

An image forming apparatus includes a plurality of container groups, and a housing part. Each container group includes a plurality of toner containers containing of toner of different colors. The housing part includes a side plate, a cover, a locking device, an upper stopper and a lower stopper. The side plate has opening rows formed for each container group in a plurality of levels in an upper-and-lower direction. The opening row includes a plurality of openings horizontally arranged. The cover is turned in an open posture for opening the opening and a close posture for closing the opening. The first stopper holds the cover at an opening angle of 90° or less to the side plate and a second stopper which holds the cover at an opening angle of larger than 90° when the cover is turned to the open posture.

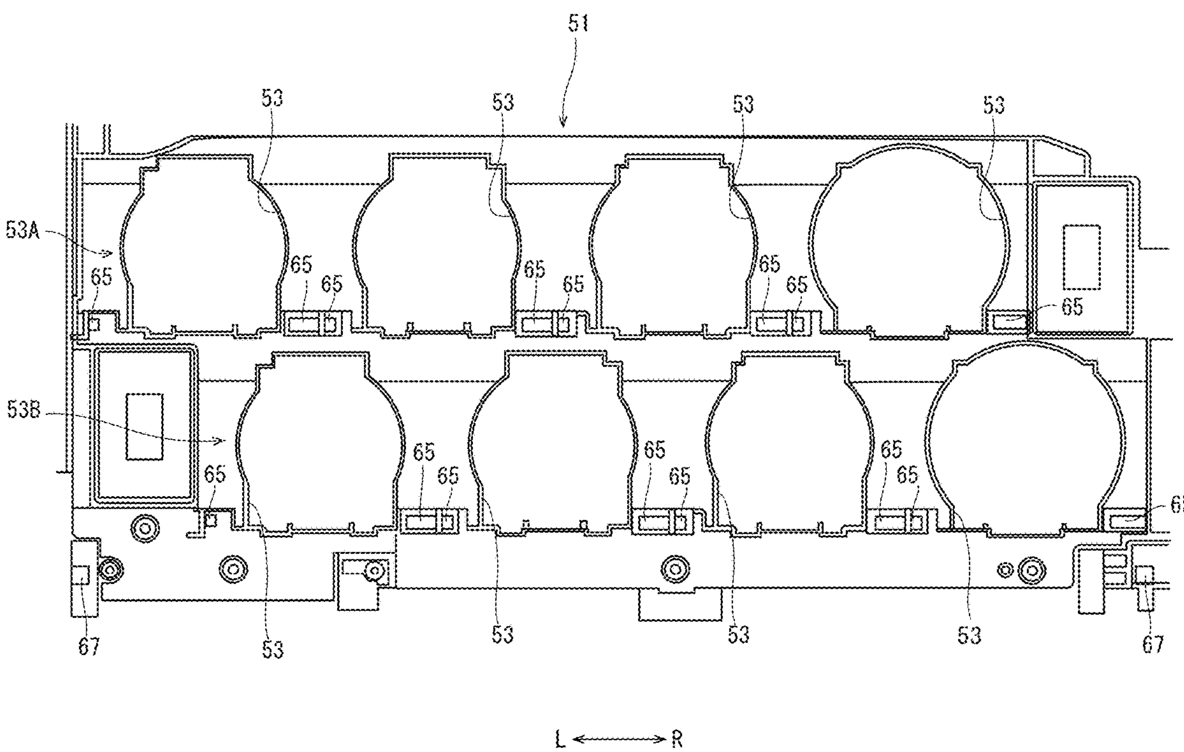

FIG. 1

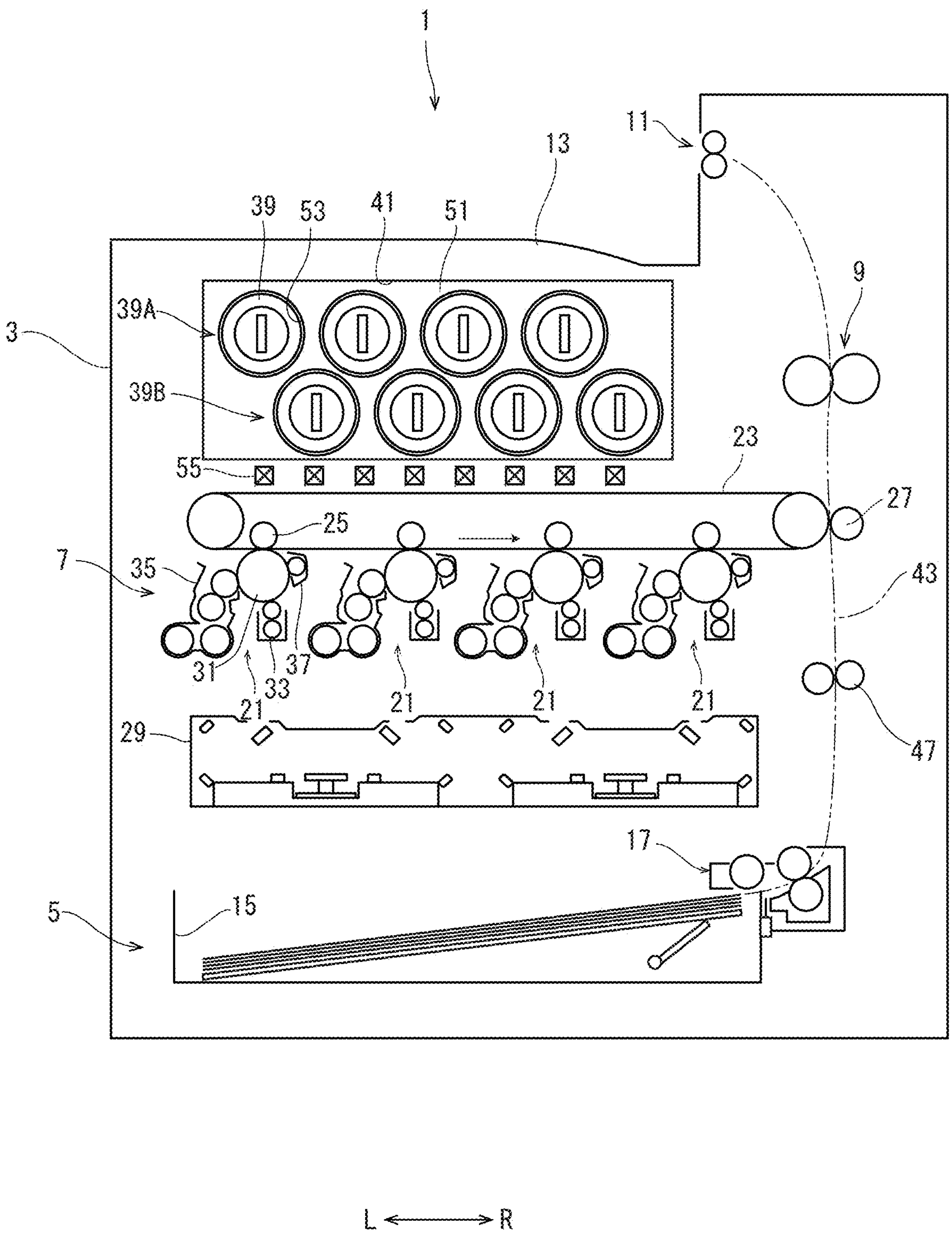
1
11
13
39
53
41
51
9
3
39A
39B
23
55
27
25
7
35
43
31
33
37
21
21
21
21
29
47
17
15
5
L
R

FIG. 2

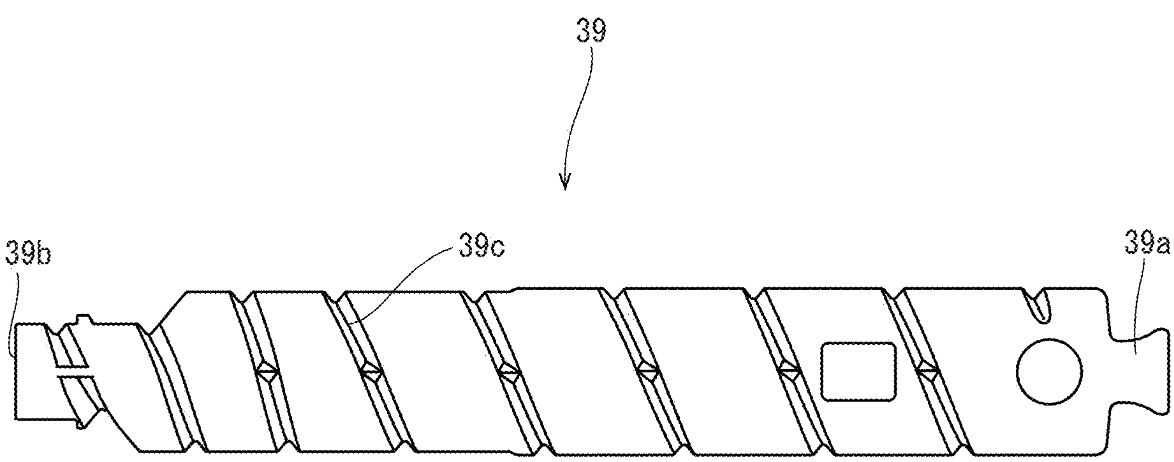

FIG. 3

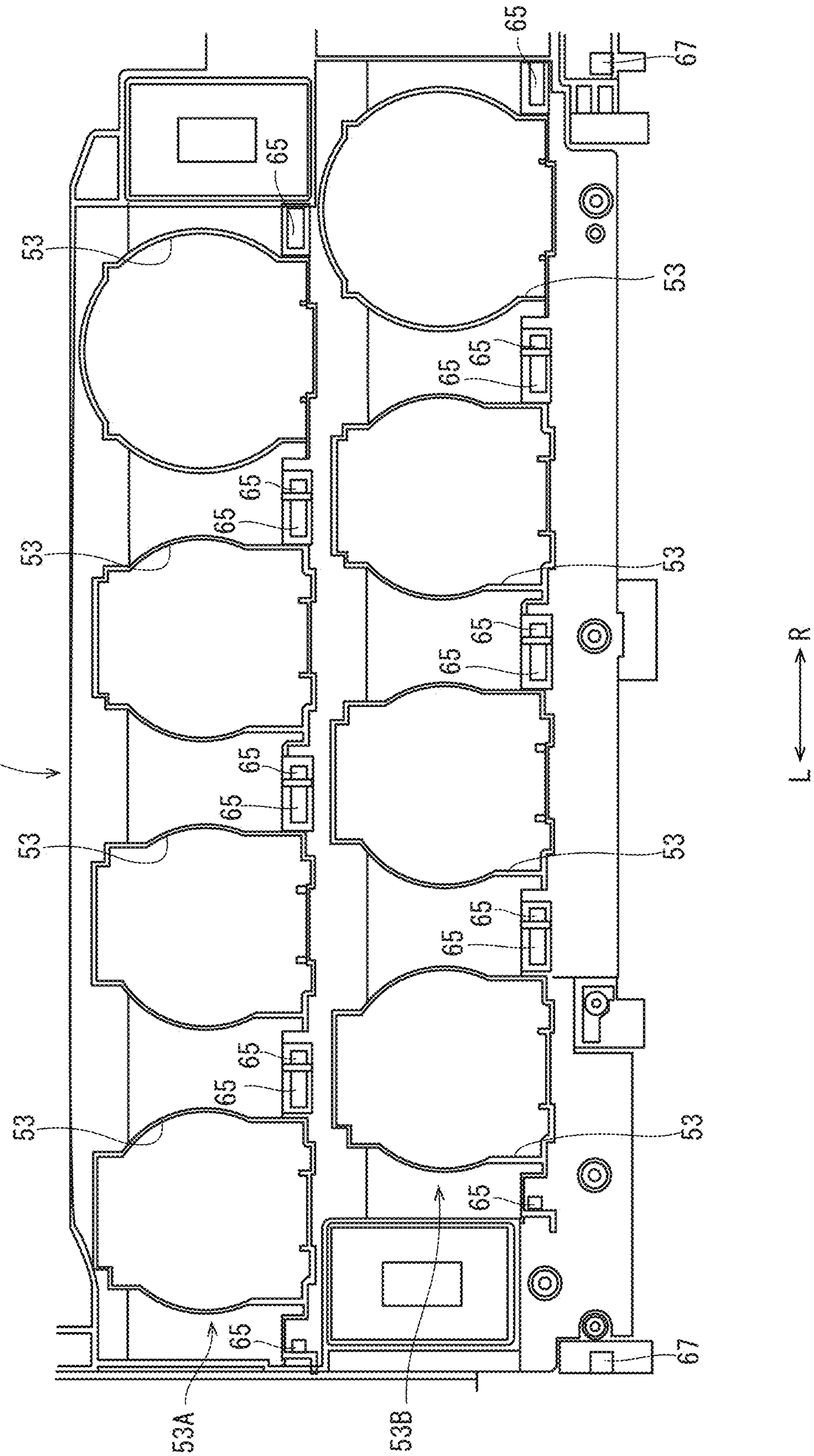

FIG. 4

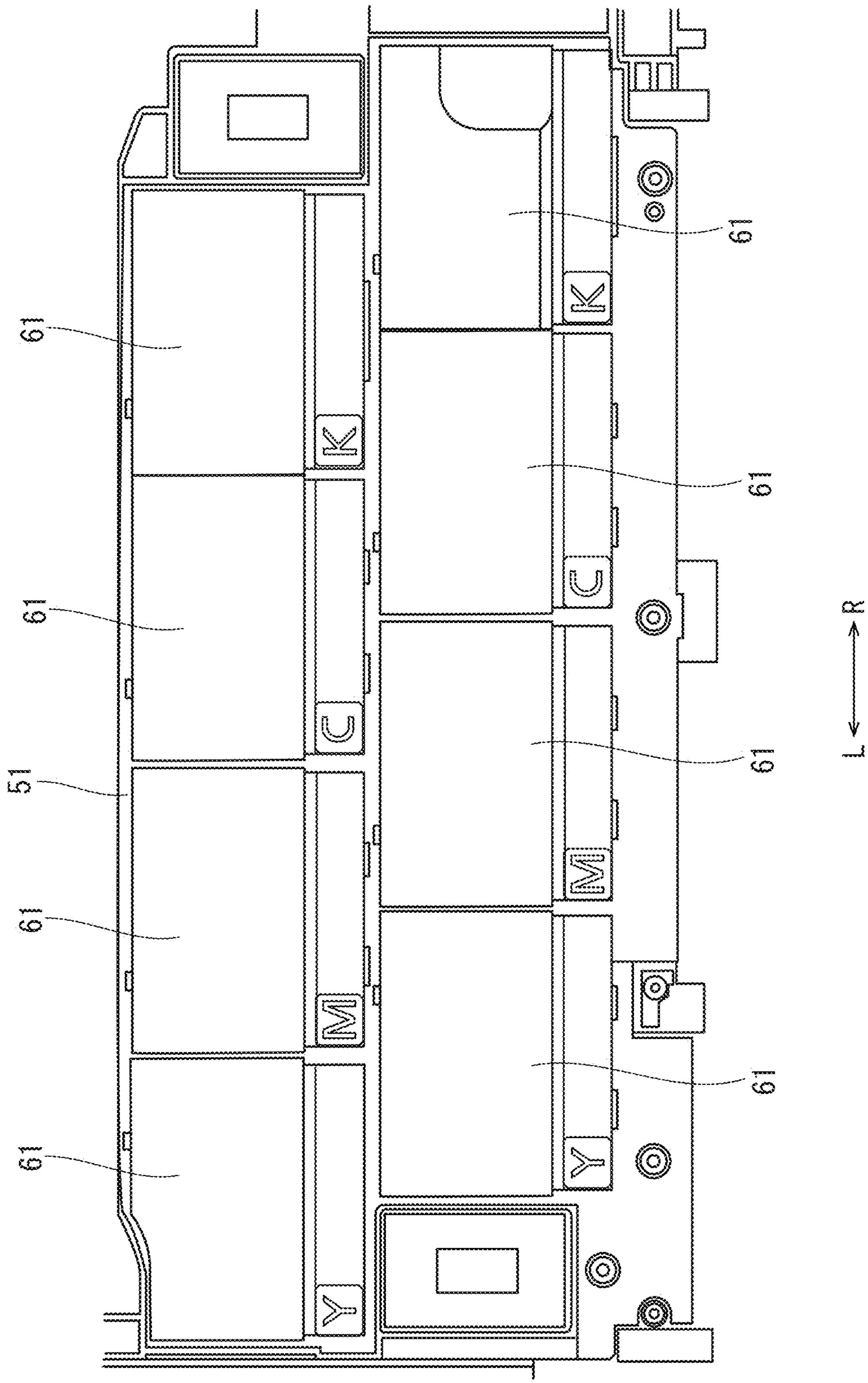

FIG. 5

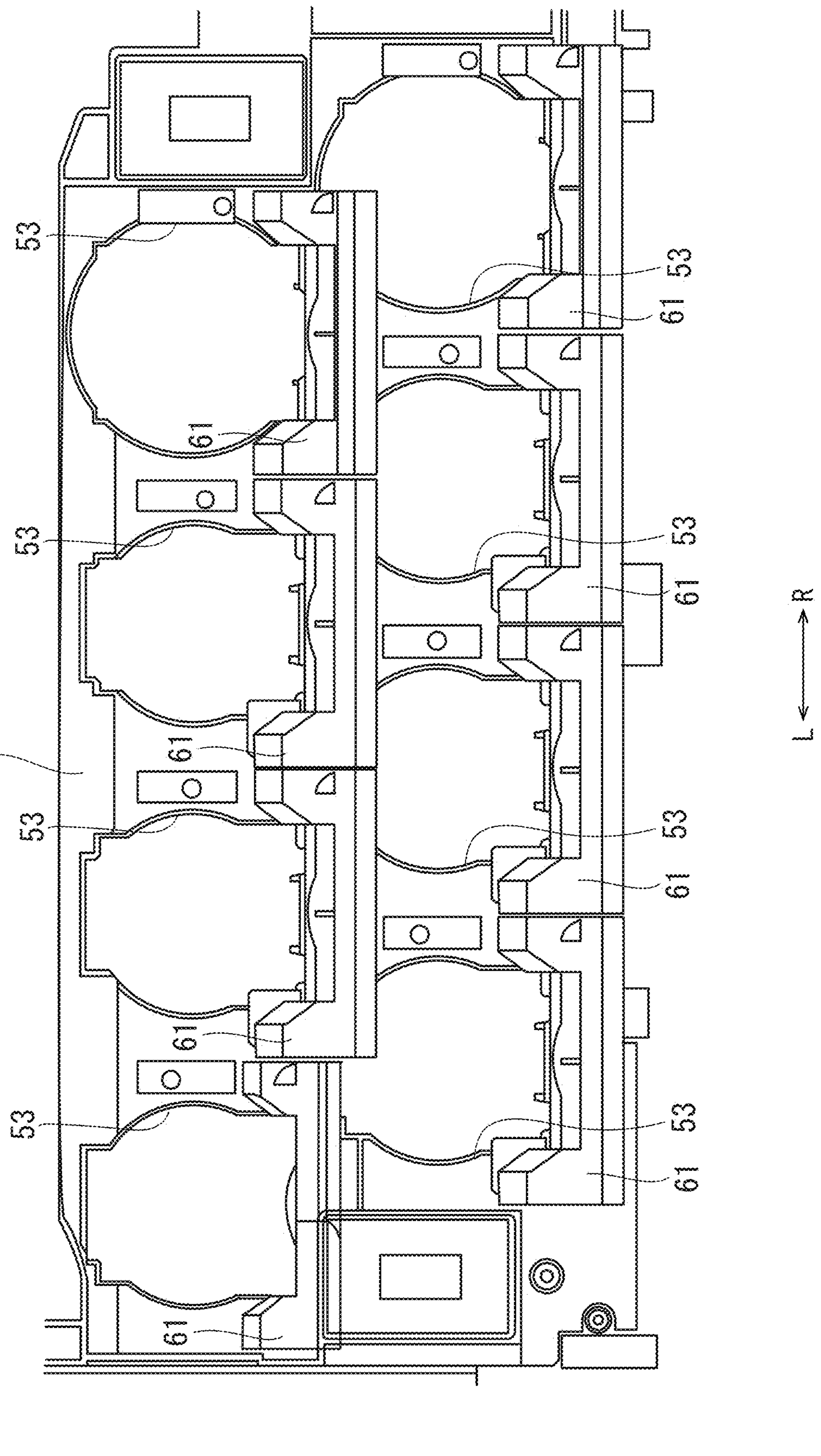

FIG. 6

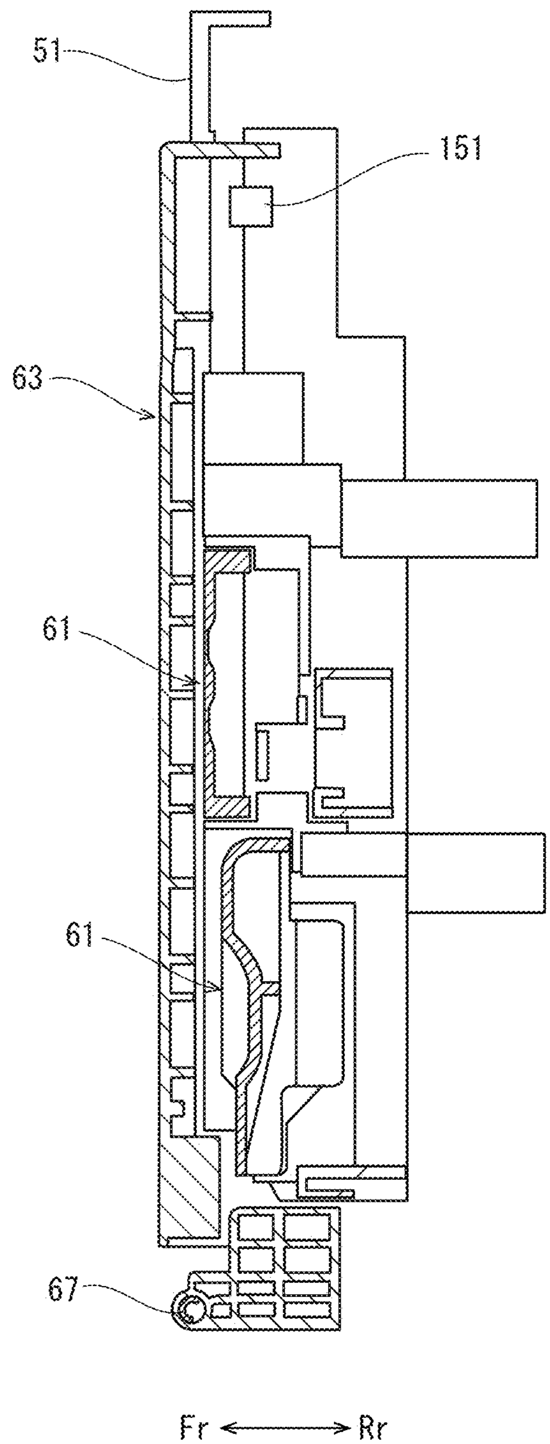

FIG. 7

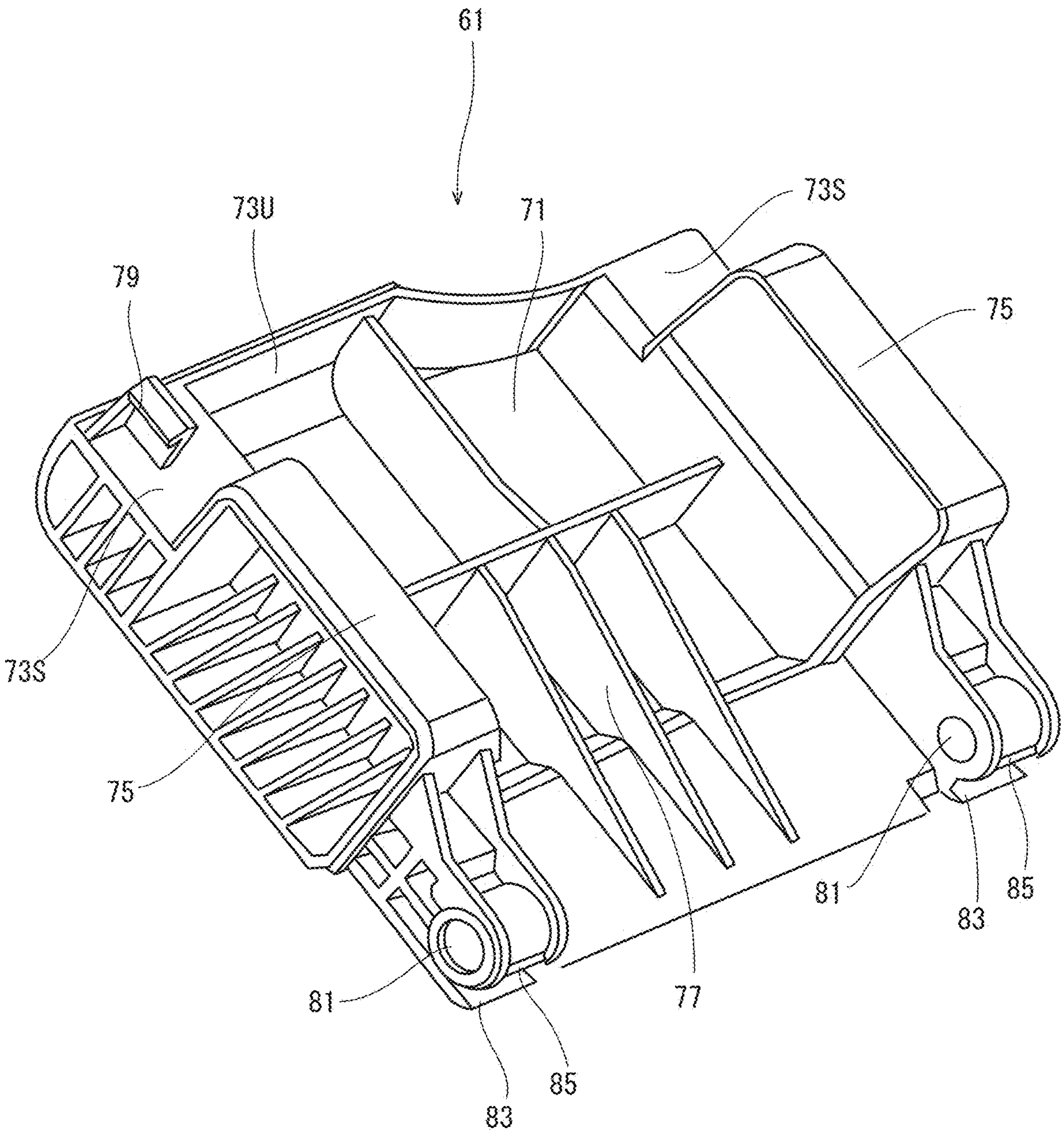

FIG. 8

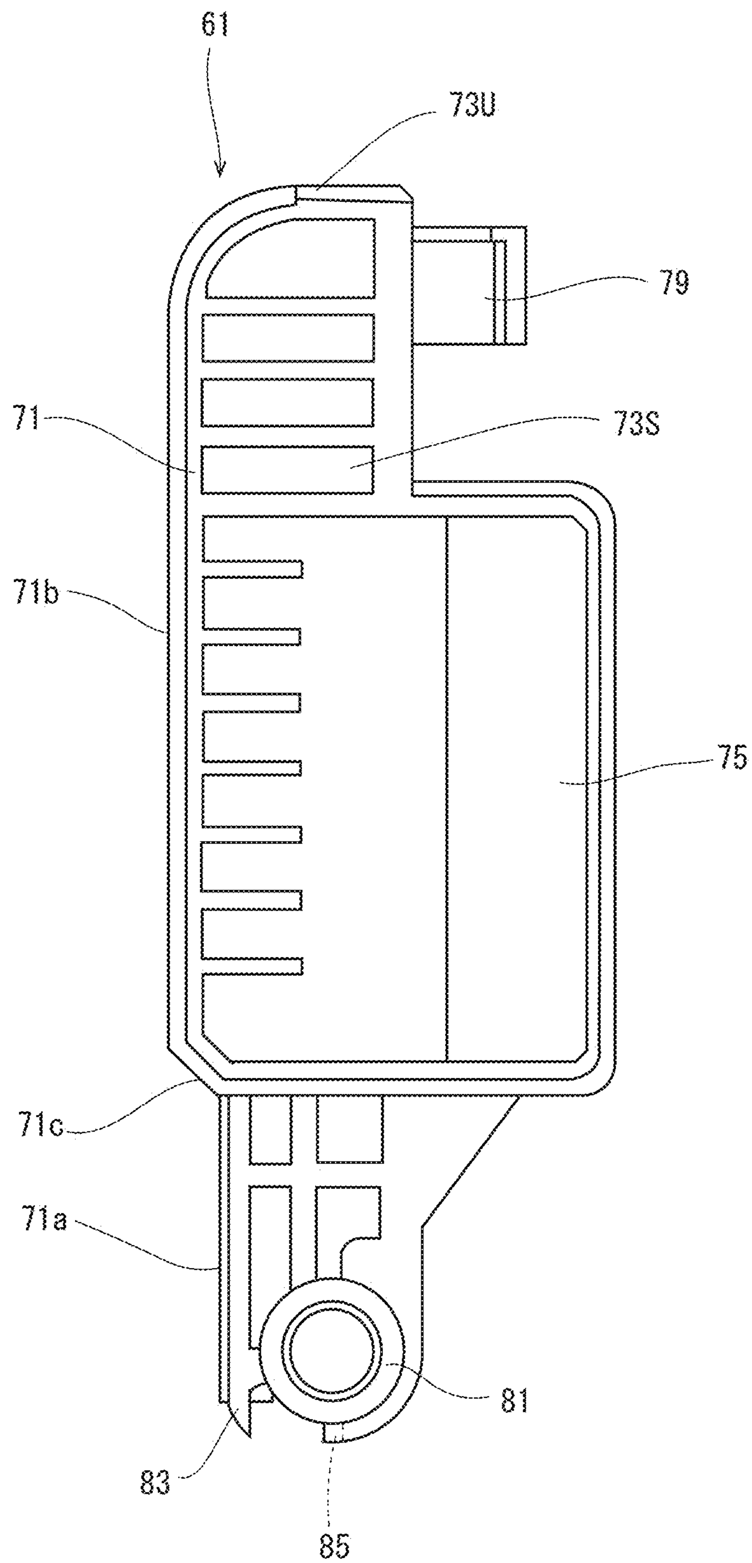

FIG. 9

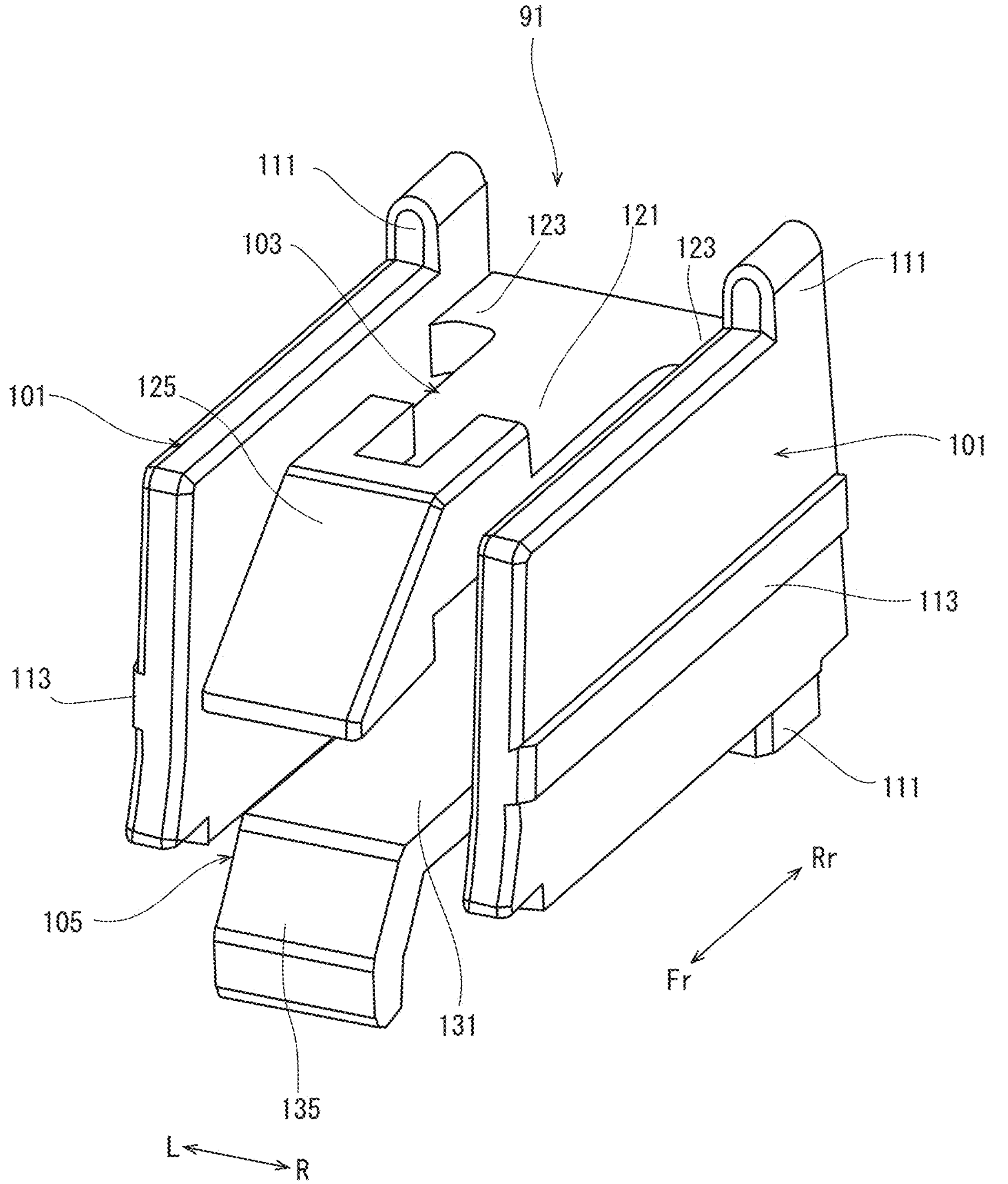
91
111
123
121
103
123
111
101
125
101
113
113
111
105
131
135
Rr
Fr
L
R

FIG. 10

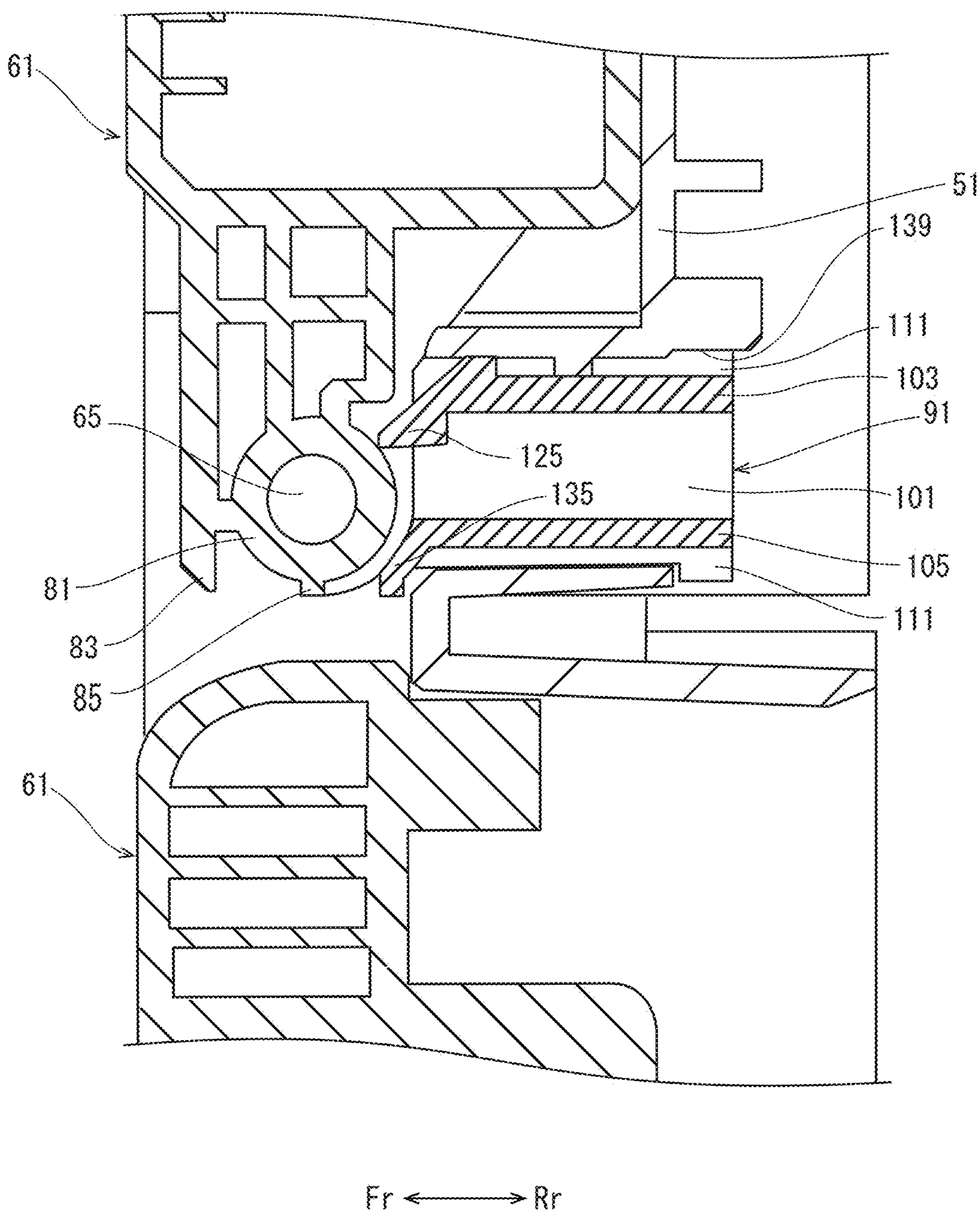

FIG. 11

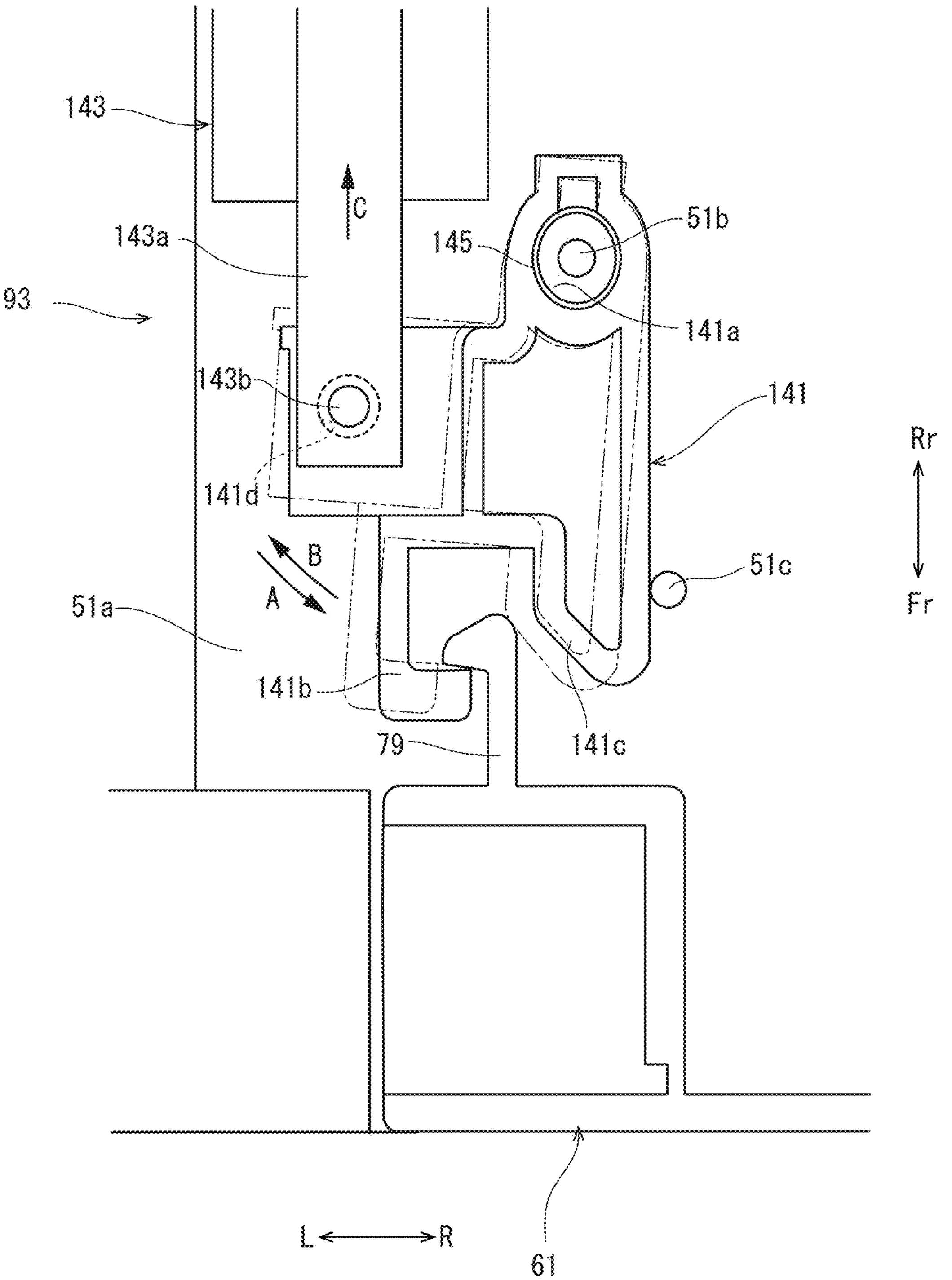
143
143a
93
143b
141d
51a
B
A
141b
79
C
145
51b
141a
141
Rr
Fr
51c
141c
L
R
61

FIG. 12

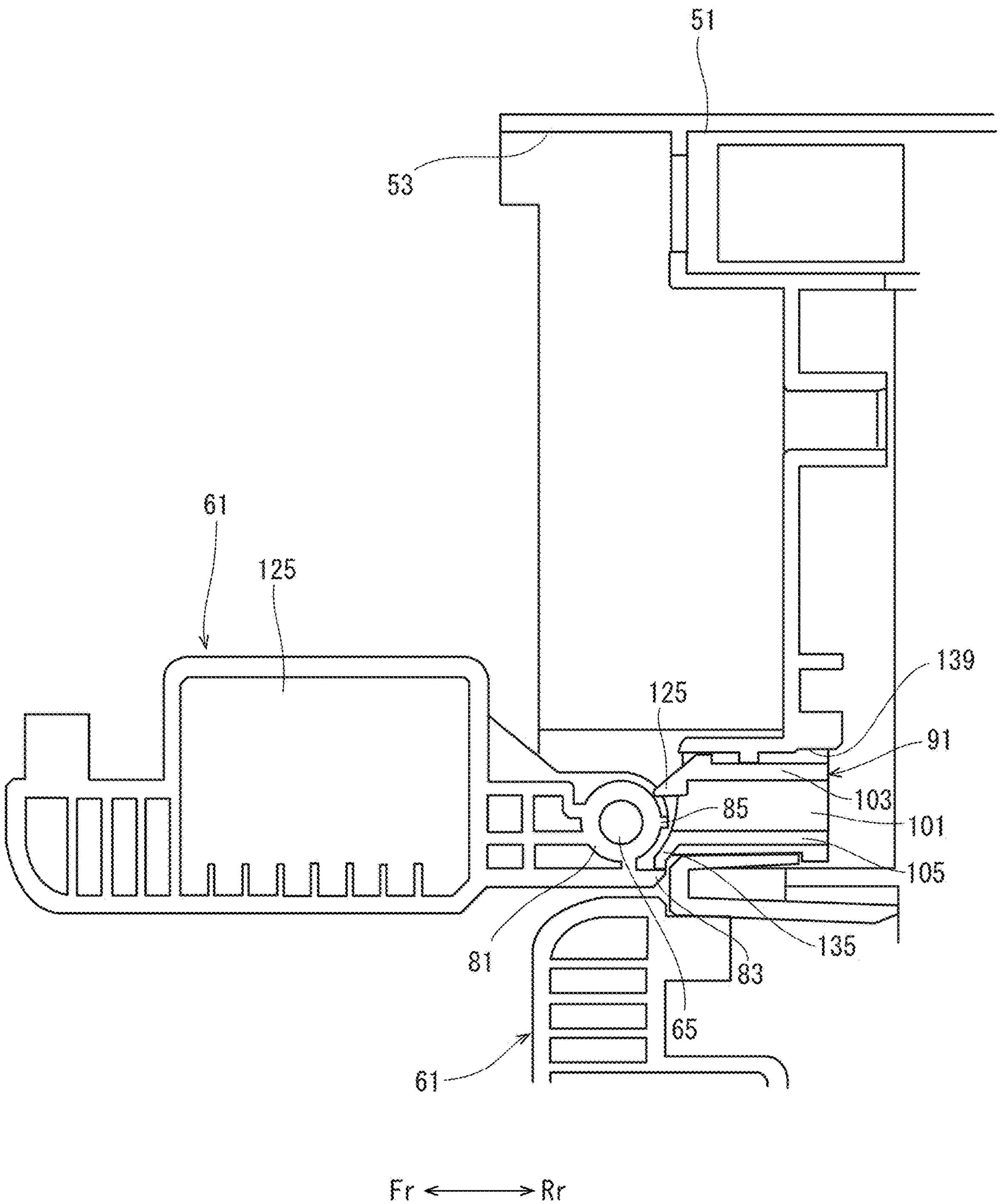

FIG. 13

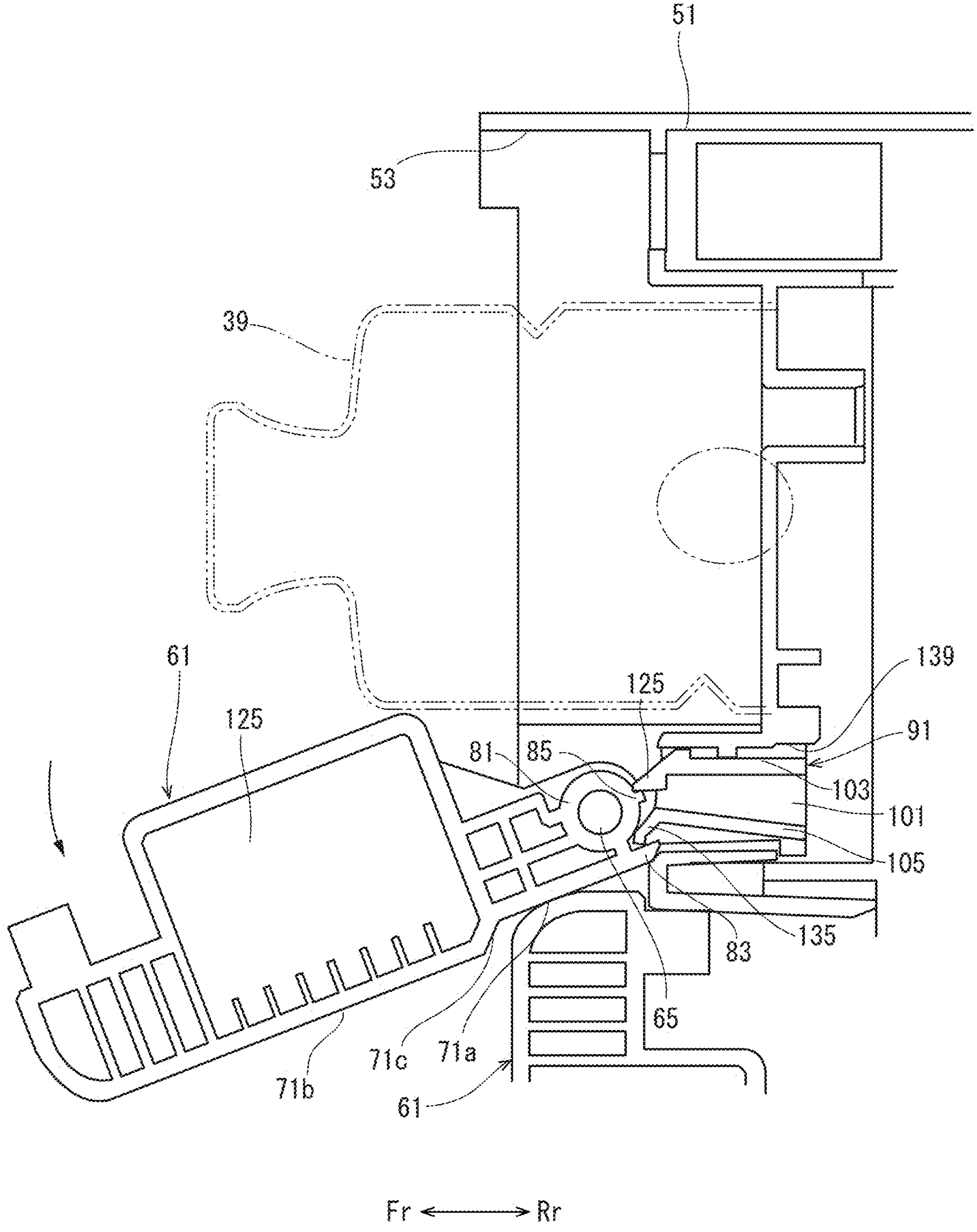
51
53
39
61
125
125
139
91
81
85
103
101
105
135
83
65
71c
71a
71b
61
Fr
Rr

IMAGE FORMING APPARATUS

INCORPORATION BY REFERENCE

[0001] This application is based on and claims the benefit of priority from Japanese patent application No.2024-029427 filed on Feb. 29, 2024, which is incorporated by reference in its entirety.

BACKGROUND

[0002] The present disclosure relates to an image forming apparatus detachable toner container.

[0003] In the conventional electrophotographic image forming apparatus, one toner container is provided for each color, and toner is supplied from each toner container to a corresponding developing device. When the toner container becomes empty, the image forming apparatus is temporarily stopped to replace the toner container.

[0004] On the other hand, the image forming apparatus may include an upper container group and a lower container group each consisting of four toner containers in which toner of four colors (yellow, cyan, magenta, and black) are contained. The upper container group and the lower container group are housed in a toner housing part in an upper row and a lower row along the left-and-right direction. Thus, two toner containers are provided for each color. Since the toner can be supplied from the other toner container even when one toner container becomes empty, the stop time of the image forming apparatus can be shortened.

[0005] In the image forming apparatus described above, a height of the apparatus main body increases to some extent in order to house the upper and lower container groups. Therefore, it is required to make the height of the apparatus main body as low as possible. When the height of the apparatus main body is lowered in this manner, a distance between an upper cover and a lower cover is also shortened. Then, a turning range of the cover overlaps with a pull-out space of the toner container, and the toner container pulled out through an opening when replacing the toner container interferes with the opened cover. Further, since a number of the toner containers is increased, it is preferable that the cover can be easily opened and closed when replacing the toner container.

SUMMARY

[0006] An image forming apparatus according to the present disclosure includes a plurality of container groups, and a housing part. Each of the container groups includes a plurality of toner containers containing of toner of different colors. To and from the housing part to and from, the toner container is attached and detached. The housing part includes a side plate, a cover, a locking device, an upper stopper and a lower stopper. The side plate has opening rows formed for each container group in a plurality of levels in an upper-and-lower direction. The opening row includes a plurality of openings through which the toner containers are attached and detached and which are arranged horizontally. The cover is supported by a rotational shaft provided at a lower end portion of each opening of the side plate, and is turned in an open posture for opening the opening and a close posture for closing the opening. The locking device locks the cover turned in the close posture to the side plate. The first stopper holds the cover at an opening angle of 90° or less to the side plate and a second stopper which holds the cover at an opening angle of larger than 90° when the cover is turned to the open posture.

[0007] The above and other objects, features, and advantages of the present disclosure will become more apparent from the following description when taken in conjunction with the accompanying drawings in which a preferred embodiment of the present disclosure is shown by way of illustrative example.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a front view schematically showing an internal structure of an image forming apparatus according to one embodiment of the present disclosure.

[0009] FIG. **2** is a side view showing a toner container in the image forming apparatus according to the embodiment of the present disclosure.

[0010] FIG. **3** is a front view showing a front side plate of the image forming apparatus according to the embodiment of the present disclosure.

[0011] FIG. **4** is a front view showing an inner cover turned in a close posture, in the image forming apparatus according to the embodiment of the present disclosure.

[0012] FIG. **5** is a front view showing the inner cover turned in an open posture, in the image forming apparatus according to the embodiment of the present disclosure.

[0013] FIG. **6** is a side cross-sectional view showing the inner cover and an outer cover which are turned in a close posture, in the image forming apparatus according to the embodiment of the present disclosure.

[0014] FIG. **7** is a perspective view showing the inner cover in the image forming apparatus according to the embodiment of the present disclosure.

[0015] FIG. **8** is a side view showing the inner cover in the image forming apparatus according to the embodiment of the present disclosure.

[0016] FIG. **9** is a perspective view showing a stopper member in the image forming apparatus according to the embodiment of the present disclosure.

[0017] FIG. **10** is a cross-sectional view showing the stopper member supported by a front side plate, in the image forming apparatus according to the embodiment of the present disclosure.

[0018] FIG. **11** is a plan view showing a locking device in the image forming apparatus according to the embodiment of the present disclosure.

[0019] FIG. **12** is a side view showing the inner cover turned in an open posture (an opening angle is) 90° in the image forming apparatus according to the embodiment of the present disclosure.

[0020] FIG. **13** is a side view showing the inner cover turned in an open posture (an opening angle is) 120° in the image forming apparatus according to the embodiment of the present disclosure.

DETAILED DESCRIPTION

[0021] Hereinafter, with the attached drawings, an image forming apparatus according to one embodiment of the present disclosure will be described.

[0022] With reference to FIG. **1**, an entire structure of the image forming apparatus **1** will be described. FIG. **1** is a front view schematically showing the internal structure of the image forming apparatus **1**. Fr, Rr, L, and R in each of the figures indicate the front side, rear side, left side, and right side of the image forming apparatus **1**, respectively.

[0023] The image forming apparatus **1** includes an apparatus main body **3** having a box-shaped housing structure. The apparatus main body **3** includes a sheet feeding part which feeds a sheet, an image forming part **7** which forms a toner image to be transferred to the sheet, a fixing part **9** which fixes the toner image to the sheet, and a discharging part **11** which discharges the sheet fixed with the toner image. A discharge tray **13** is provided on the upper surface of the apparatus main body **3**.

[0024] The sheet feeding part **5** is disposed in the lower portion of the apparatus main body **3**. The sheet feeding part **5** includes a sheet feeding cassette **15** in which the sheets are stored, and a sheet feeding device **17** which feeds the sheet stored in the sheet feeding cassette **15**.

[0025] The image forming part **7** is disposed above the sheet feeding part **5**. The image forming part **7** includes four image forming units **21** corresponding to toner of four colors (yellow, cyan, magenta, and black), an intermediate transfer belt **23**, primary transfer rollers **25** corresponding to the four image forming units **21**, a secondary transfer roller **27**, and an exposure device **29**. Each image forming unit **21** includes a rotatable photosensitive drum **31**, and a charging device **33**, a developing device **35**, and a cleaning device **37** which are disposed around the photosensitive drum **31**. The four image forming units **21** are arranged side by side in the left-and-right direction. Each developing device **35** is connected to a toner container **39** storing the toner of the corresponding color.

[0026] The toner containers **39** are housed in a container housing part **41** provided above the image forming part **7**. The toner container **39** and the container housing part **41** will be described later.

[0027] The intermediate transfer belt **23** is disposed above the four image forming units **21**, and wound between a driving roller and a driven roller which are disposed apart in the left-and-right direction, and travels in the direction indicated by the arrow in FIG. **1**. The outer surface of the lower track of the intermediate transfer belt **23** faces the photosensitive drums **31** of the four image forming units **21**, and a primary transfer nip is formed between the photosensitive drum **31** and the intermediate transfer belt **23**. The four primary transfer rollers **25** are disposed in the hollow space of the intermediate transfer belt **23**, and face the photosensitive drums **31** of the four image forming units **21** across the intermediate transfer belt **23**. The secondary transfer roller **27** faces the driving roller across the intermediate transfer belt **23**, and a secondary transfer nip is formed between the secondary transfer roller **27** and the intermediate transfer belt **23**.

[0028] The fixing part **9** is disposed above the image forming part **7**. The fixing part **9** includes a heating roller and a pressing roller, and a fixing nip is formed between both rollers. The discharging part **11** is disposed above the fixing part **9**. The discharging part **11** includes a discharge rollers pair.

[0029] Further, a sheet conveyance path **43** is formed in the apparatus main body **3** from the sheet feeding device **17** of the sheet feeding part **5** to the discharge rollers pair of the discharging part **11** through the secondary transfer nip and the fixing nip. On the conveyance path **43**, a registration rollers pair **47** is provided on the upstream side of the secondary transfer nip in the sheet conveyance direction.

[0030] The image forming operation will be briefly described. In each image forming unit **21** of the image forming part **7**, the charging device **33** charges the photosensitive drum **31** to a predetermined potential. Thereafter, the photosensitive drum **31** is exposed by the exposure device **29** based on image data to form an electrostatic latent image. The electrostatic latent image is developed into a toner image by the developing device **35**. At the primary transfer nip, the toner image is transferred from the photosensitive drum **31** to the intermediate transfer belt **23** by the primary transfer roller **25** to which the transfer bias is applied. In each of the four image forming units **21**, the toner images are transferred to the intermediate transfer belt **23** to form a full-color toner image on the intermediate transfer belt **23**.

[0031] On the other hand, in the sheet feeding part **5**, the sheet is fed from the sheet feeding cassette **15** to the conveyance path **43** by the sheet feeding device **17**. The sheet is conveyed along the conveyance path **43**, and after the skew is corrected by the registration rollers pair **47**, the sheet is conveyed to the secondary transfer nip. At the secondary transfer nip, the full-color toner image is transferred from the intermediate transfer belt **23** to the sheet by the secondary transfer roller **27** to which the transfer bias is applied. Thereafter, the sheet is conveyed to the fixing part **9**, and the toner image is heated and pressurized at the fixing nip to be fixed to the sheet. The sheet is conveyed to the discharging part **11**, and discharged to the discharge tray **13** by the discharge rollers pair.

[0032] Next, the toner container **39** will be described with reference to FIG. **2**. FIG. **2** is a side view showing the toner container **39**. The toner container **39** is formed in a cylindrical shape having a hollow space for storing the toner. A small diameter projection **39***a* is provided on one end surface in the longitudinal direction of the toner container **39**, and a discharge port **39***b* is provided on the other end surface in the longitudinal direction. On the inner circumferential surface of the toner container **39**, a spiral protrusion **39***c* is formed along the longitudinal direction.

[0033] The toner containers **39** for storing the yellow, cyan and magenta toners have the same shape and the same capacity, and the toner container **39** for storing the black toner has a larger capacity than these toner containers **39**. A memory chip (not shown) storing various kinds of information is stuck to each toner container **39**.

[0034] Next, the container housing part **41** will be described with reference to FIG. **1** and FIG. **3** to FIG. **6**. FIG. **3** is a front view showing the front side plate **51** of the apparatus main body **3**, FIG. **4** is a front view showing inner covers **61** turned in a close posture, FIG. **5** is a front view showing the inner covers **61** turned in an open posture, and FIG. **6** is a side view showing the inner cover **61** and the outer cover **63** turned in a close posture.

[0035] As shown in FIG. **1**, the container housing part **41** houses an upper container group **39**A and a lower container group **39**B each consisting of four toner containers **39** containing the toner **4** four colors (yellow, cyan, magenta, and black). That is, two toner containers **39** are provided for each color. Thus, as described above, even when one toner container **39** becomes empty, the toner can be supplied from the other toner container **39**, so that the stop time of the image forming apparatus **1** associated with the replacement of the toner container can be shortened.

[0036] The container housing part **41** is formed between a front side plate **51** and a rear side plate (not shown) of the apparatus main body **3**. As shown in FIG. **3**, the front side plate **51** has upper and lower opening rows **53**A and **53**B corresponding to the upper and lower container groups **39**A and **39**B. Each of the opening rows includes four openings **53** arranged in a horizontal direction (the left-and-right direction). The respective openings **53** of the upper opening row **53**A and the respective openings **53** of the lower opening row **53**B are displaced in the left-and-right direction by a distance between the adjacent openings **53** in the left-and-right direction. The toner container **39** is attached to and detached from the container housing part **41** through the opening **53**.

[0037] In each opening row, the four openings **53** correspond to the toner containers **39** of yellow, magenta, cyan, and black in order from the left. The openings **53** corresponding to yellow, magenta and cyan have substantially the same size and shape, and the opening **53** corresponding to black is larger than these openings **53**.

[0038] The container housing part **41** is provided with a toner replenishing mechanism (not shown) which replenishes the toner from the toner container **39** to the corresponding developing device **35** for each opening **53**. The toner replenishing mechanism includes a rotating mechanism for rotating the toner container **39**, a pipe arranged between the toner container **39** and the corresponding developing device **35** along the upper-an-lower direction, and a toner sensor **55** (see FIG. **1**) for detecting that the toner container **39** becomes empty. The pipe is made of transparent material. As described above, since the openings **53** of each opening row are displaced in the left-and-right direction, the pipe corresponding to the upper opening **53** can be disposed between the pipes of the toner supply mechanisms corresponding to the adjacent openings **53** of the lower opening row.

[0039] The toner sensor **55** is a permeability sensor, and detects the presence or absence of the toner in the pipe. The toner sensor **55** is electrically connected to a control part (not shown). A detection result of the toner sensor **55** is transmitted to the control part.

[0040] The front side plate **51** rotatably supports an inner cover **61** (see FIG. **4** and FIG. **5**) which opens and closes each opening **53**, and an outer cover **63** (see FIG. **6**) which covers the outside of all the inner covers **61**.

[0041] As shown in FIG. **3**, right and left inner rotational shafts **65** for rotatably supporting the inner cover **61** are provided on both left and right sides of the lower end portion of each opening **53**. The inner cover **61** is rotatable around the right and left inner rotational shafts **65** in a close posture for closing the opening **53** (see FIG. **4**) and an open posture for opening the opening **53** (see FIG. **5**). Further, as shown in FIG. **3**, left and right outer rotational shafts **67** for rotatably supporting the outer cover **63** are provided at the right and left lower corners of the front side plate **51**. The outer cover **63** is rotatable around the right and left outer rotational shafts **67** in a close posture (see FIG. **6**) for covering the outside of all the inner covers **61** and an open posture for exposing all the inner covers **61**.

[0042] Next, the inner cover **61** will be described with reference to FIG. **7** and FIG. **8**. FIG. **7** is a perspective view showing the inner cover **61**, and FIG. **8** is a side view showing the inner cover **61**.

[0043] All the inner covers **61** have the same size (a length in the upper-and-lower direction, and a length in the left-and-right direction) and structure (see FIG. **4**). However, depending on the position of the inner cover **61**, cutouts or recesses may be provided in order to avoid interference with other components. For example, a notch is formed in the upper left corner of the inner cover **61** for opening and closing the yellow toner container opening **53** in the upper row, and a recess is formed in the inner cover **61** for opening and closing the black toner container opening **53** in the lower row.

[0044] As shown in FIG. **7** and FIG. **8**, the inner cover **61** has a rectangular cover plate **71** having a predetermined height (a length in the upper-and-lower direction) and width (a length in the left-and-right direction). A mark indicating the color of the corresponding toner is marked on the outer surface of the cover plate **71**. As shown in FIG. **8**, the cover plate **71** is formed so that a lower portion **71***a* is recessed with respect to an upper portion **71***b* through an inclined portion **71***c* in the thickness direction (the front-and-rear direction).

[0045] As shown in FIG. **7**, on the inner surface of the cover plate **71**, an upper wall **73**U is protruded in the thickness direction along the upper edge, and left and right walls **73**S are protruded in the thickness direction along the left and right side edges. The right and left walls **73**S have a predetermined width in the left-and-right direction. In the center portions of the left and right walls **73**S, protruding portions **75** protruding further in the thickness direction are formed. As shown in FIG. **8**, the upper outer corners of the cover plate **71** and the upper and lower outer corners of the protruding portion **75** are rounded into the R surface.

[0046] As shown in FIG. **7**, on the inner surface of the cover plate **71**, a plurality of ribs **77** along the upper-and-lower direction and in the left-and-right direction are provided in the space between the upper wall **73**U and the left and right walls **73**S. Further, at one upper end portion of the left and right walls **73**S, a hook **79** is provided which is engaged with the front side plate **51** by a locking device **93** (described later with reference to FIG. **11**).

[0047] At the lower end portions of the left and right walls **73**S, cylindrical right and left bearing portions **81** fitted to the right and left inner rotational shafts **65** of the front side plate **51** are provided. A first protrusion **83** and a second protrusion **85** are formed on the outer circumferential surface of each bearing portion **81**. The first protrusion **83** and the second protrusion **85** are spaced apart in the circumferential direction. In this example, the first protrusion **83** and the second protrusion **85** are arranged at a center angle of approximately 90° with respect to the center of the bearing portion **81**. The first protrusion **83** and the second protrusion **85** have a predetermined width along the left-and-right direction. The first protrusion **83** has a tapered claw shape extending in a direction parallel to the tangential direction of the bearing portion **81** along an extension of the outer surface of the lower portion **71***a* of the cover plate **71**. The second protrusion **85** is disposed inside the first protrusion **83**, and protrudes in the radial direction of the bearing portion **81**. A height of the second protrusion **85** (a length along the radial direction from the outer surface of the bearing portion **81**) is lower than a height of the first protrusion **83**.

[0048] With the above configuration, the center of gravity of the inner cover **61** in the close posture is positioned on the outer side (on the front side) of the front side plate **51** than the right and left inner rotational shafts **65**. Thus, the inner cover **61** can be automatically turned to the open posture.

[0049] Further, the front side plate **51** is provided with a stopper member **91** which holds the inner cover **61** turned in the open posture at at least two opening angles with respect to the front side plate **51**, and a locking device **93** which locks the inner cover **61** turned in the close position to the front side plate **51**.

[0050] First, the stopper member **91** will be described with reference to FIG. **9** and FIG. **10**. FIG. **9** is a perspective view showing the stopper member **91**, and FIG. **10** is a side view showing the stopper member **91** supported by the front side plate **51**. The stopper member **91** holds the inner cover **61** at an opening angle of 90° with respect to the front side plate **51** and at an opening angle of 120° larger than 90°. The stopper member **91** has right and left side plates **101** facing each other in the left-and-right direction, and upper and lower stopper pieces **103** and **105** cantilevered by the right and left side plates **101**. The stopper member **91** is made of, for example, resin.

[0051] The right and left side plates **101** have a rectangular shape when viewed from the left-and-right direction. Upper and lower protrusions **111** are formed on the rear end portion of the upper end surface and on the rear end portion of the lower end surface of each side plate **101**. On the outer surface of each side plate **101**, an outer protrusion **113** is formed along the horizontal direction.

[0052] The upper stopper piece **103** has a body portion **121**, right and left fixing portion **123**, and an engagement portion **125**. The body portion **121** is a rectangular plate piece having a predetermined width along the left-and-right direction, a predetermined length along the front-and-rear direction, and a predetermined thickness. The right and left fixing portions **123** protrude in opposite directions from the rear end portions of the right and left side surfaces of the body portion **121**. The right and left fixing portions **123** are fixed to the rear end portions of the inner surfaces of the right and left side plates **101**.

[0053] The engagement portion **125** is provided at the front end portion of the main body portion **121**. The engagement portion **125** is thicker than the body portion **121**, and is formed in a trapezoidal shape when viewed from the left-and-right direction. The engagement portion **125** has flat upper and lower surfaces, a front end surface inclined downward, and a rear end surface along the upper-and-lower direction. The front end of the engagement portion **125** protrudes forward from the left and right side plates **101**. The upper surfaces of the engagement portion **125** are formed at the same height as the upper surfaces of the left and right side plates **101**.

[0054] The lower stopper piece **105** has a body portion **131**, right and left fixing portions, and an engagement portion **135**. The main body portion **131** is a rectangular plate piece having the same width, length and thickness as the main body portion **121** of the upper stopper piece **103**. The right and left fixing portions protrude in opposite directions from the rear end portions of the right and left side surfaces of the main body portion **131**. The right and left fixing portions are fixed to the rear end portions of the inner surfaces of the right and left side plates **101** below the upper stopper piece **103**. That is, a space is formed above the lower stopper piece **105**. The engagement portion **135** has the same thickness as the main body portion **131**, and has an inclined portion bent diagonally downward from the front end portion of the main body portion **131** and a bent portion bent downward from the front end portion of the inclined portion. The engagement portion **135** protrudes forward and downward from the right and left side plates **101**.

[0055] As shown in FIG. **10**, the stopper member **91** is supported by a housing space **139** formed in the front side plate **51** of the apparatus main body **3**. The housing space **139** is a substantially rectangular parallelepiped space formed on the rear side of the left and right inner rotational shafts **65**, and surrounded by left and right wall surfaces and upper and lower wall surfaces. The stopper member **91** is inserted into the housing space **139** from the rear side of the front side plate **51**, and the upper and lower protrusions **111** of the right and left side plates **101** are brought into contact with the front side plate **51**, whereby the stopper member **91** is supported by the housing space **139**. The outer protrusions **113** of the right and left side plates **101** are brought into contact with the right and left wall surfaces of the housing space **139**.

[0056] The front end portion of the engagement portion **125** of the upper stopper piece **103** protrudes forward from the housing space **139**. The engagement portion **135** of the lower stopper piece **105** protrudes forward and downward from the housing space **139**. The protruded front end portion of the engagement portion **125** of the upper stopper piece **103** and the protruded engagement portion **135** of the lower stopper piece **105** are spaced apart in the circumferential direction of the bearing portion **81** around the bearing portion **81** of the inner cover **61** fitted to the inner rotational shafts **65**. A predetermined interval is formed between the engagement portion **135** of the lower stopper piece **105** and the bearing portion **81**. The front end portion of the engagement portion **125** of the upper stopper piece **103** is substantially in contact with the bearing portion **81**.

[0057] The upper surface of the engagement portion **125** of the upper stopper piece **103** is in contact with the upper surface of the housing space **139**. Thus, an upward turning of the upper stopper piece **103** is restricted. On the other hand, since a space is formed above the lower stopper piece **105**, the lower stopper piece **105** can be elastically turned upward around the right and left fixing portions.

[0058] Next, the locking device **93** will be described with reference to FIG. **11**. FIG. **11** is a plan view showing the locking device **93**. The locking device **93** is provided on the front side plate **51**, and is engaged with the hook **79** provided on the inner cover **61** to lock the inner cover **61** turned in the close posture to the front side plate **51**. The locking device **93** is supported by a horizontal fixing plate **51*a*** provided on the front side plate **51** corresponding to each opening **53**. The locking device **93** includes a hook member **141** capable of being engaged with the hook **79** of the inner cover **61** by turning, and a solenoid **143** for turning the hook member **141**.

[0059] The hook member **141** has a base end portion provided with a shaft hole **141*a***, and a distal end portion provided with a hook piece **141*b*** and a pressing piece **141*c***. A rotational shaft **51*b*** provided in the fixing plate **51*a*** is inserted into the shaft hole **141*a***. As a result, the hook member **141** is turned on the fixing plate **51*a*** around the rotational shaft **51*b***. The hook piece **141*b*** and the pressing piece **141*c*** are arranged at a predetermined interval along the rotational locus around the shaft hole **141*a***. The hook member **141** is biased by a torsion coil spring **145** disposed in the shaft hole **141***a* so that the hook piece **141***b* is turned in an engagement direction A in which the hook piece **141***b* is engaged with the hook **79** of the inner cover **61**. The turning the hook piece **141***b* in the engagement direction A is restricted by a pin **51***c* provided on the fixing plate **51***a*.

[0060] When the hook member **141** is biased by the torsion coil spring **145** and turned in the engagement direction A, the hook piece **141***b* can be engaged with the hook **79** of the inner cover **61**. On the other hand, when the hook member **141** is turned in a release direction B, which is opposite to the engagement direction A (see the two-dot chain line in FIG. **11**), the engagement between the hook piece **141***b* and the hook **79** is released, and the hook **79** is pressed by the pressing piece **141***c* so that the inner cover **61** is turned in the open posture.

[0061] The solenoid **143** has a movable member **143***a* movable forward and rearward along the front-and-rear direction. The distal end of the movable member **143***a* is rotatably connected to the hook member **141**. Specifically, the distal end of the movable member **143***a* has a shaft **143***b*, and the hook member **141** has an opening **141***d* into which the shaft **143***b* is inserted. A diameter of the opening **141***d* is larger than a diameter of the shaft **143***b*. As the movable member **143***a* moves forward and rearward, the hook member **141** is turned around the rotational shaft **51***b* via the shaft **143***b* and the opening **141***d*. Specifically, when the movable member **143***a* is moved forward, the hook member **141** is turned in the engagement direction A, when the movable member **143***a* is moved rearward, the hook member **141** is turned in the release direction B.

[0062] The solenoid **143** is electrically connected to the control part. The solenoid **143** is switched between an energized state and a non-energized state by the control part.

[0063] Next, the outer cover **63** will be described with reference to FIG. **6**. The outer cover **63** has a rectangular shape having substantially the same size as the front side plate **51**. On the inner surface of the outer cover **63**, a plurality of ribs are provided along the upper-and-lower direction and the left-and-right direction. The outer cover **63** is rotatably supported by the right and left outer rotational shafts **67** provided on the front side plate **51**. The outer cover **63** is turned in a close posture for covering the front side plate **51** and an open posture for opening the front side plate **51**.

[0064] A locking device (not shown) locks the outer cover turned in the open posture to the front side plate **51**. An opening angle of the outer cover **63** with respect to the front side plate **51** turned in the open posture is approximately 90°. The front side plate **51** is provided with an opening/closing detection switch **151** for detecting the opening/closing of the outer cover **63**. The opening/closing detection switch **151** is switched from ON to OFF when the outer cover **63** is slightly opened from the position of FIG. **6** (the close posture). The opening/closing detection switch **151** is electrically connected to the control part.

[0065] The opening and closing operation of the inner cover **61** in the image forming apparatus **1** having the above configuration will be described with reference to FIG. **12** and FIG. **13**. FIG. **12** is a side view showing the inner cover **61** turned in the open posture (an opening angle is) 90°, and FIG. **13** is a side view showing the inner cover **61** turned in the open posture (an opening angle is) 120°.

[0066] During the image forming operation, the outer cover **63** and the inner cover **61** are turned to the close posture, and locked to the close posture by the respective locking devices. When the inner cover **61** is turned in the close posture, the upper wall **73**U, the right and left walls **73**S and the protruding portions **75** abut against the front side plate **51** on both sides of the opening **53**, and the opening **53** is closed by the cover plate **71**. Further, as shown in FIG. **4**, when all the inner covers **61** are turned in the close posture, there is almost no gap between the adjacent inner covers **61** in the left-and-right direction, and a gap between the inner covers **61** in the upper-and-lower direction is narrowed as much as possible.

[0067] When the control part determines that the predetermined toner container **39** becomes empty based on the detection result of the toner sensor **55**, a message indicating that the toner container **39** needs to be replaced is displayed on a display part (not shown) or the like. After confirming the message, the user turns the outer cover **63** to the open posture. Then, the opening/closing detection switch **151** is switched from ON to OFF, and the control part switches the solenoid **143** of the locking device **93** provided in the corresponding opening **53** from the non-energized state to the energized state only for a predetermined period of time.

[0068] As a result, the movable member **143***a* is moved rearward (see the arrow C in FIG. **11**), and the hook member **141** is turned in the release direction B to separate from the hook **79** of the inner cover **61**. Further, the pressing piece **141***c* of the hook member **141** presses the hook **79**. As described above, in the close posture of the inner cover **61**, the center of gravity of the inner cover **61** is positioned on the outer side of the front side plate **51** than the right and left inner rotational shafts **65**. In addition to the above configuration, the pressing piece **141***c* of the hook member **141** pushes the hook **79**, so that when the toner container **39** becomes empty, the corresponding inner cover **61** is automatically and surely turned to the open posture.

[0069] Therefore, when the outer cover **63** is opened, the inner cover **61** is automatically turned in the open posture while leaning on the outer cover **63**.

[0070] As shown in FIG. **12**, the inner cover **61** is turned until the first protrusion **83** formed in the bearing portion **81** is engaged with the engagement portion **135** of the lower stopper piece **105** of the stopper member **91**. At this time, an opening angle of the inner cover **61** with respect to the front side plate **51** is 90° in this example. As described above, the lower stopper piece **105** is an example of the first stopper which holds the inner cover **61** at an opening angle of 90° or less in the present disclosure. Since a height of the second protrusion **85** is lower than a height of the first protrusion **83**, the second protrusion **85** passes between the lower stopper piece **105** and the bearing portion **81** during the turning of the inner cover **61**.

[0071] When the inner cover **61** is turned in the open posture in the above manner, the opening **53** of the front side plate **51** is exposed. However, in this state, the protruding portions **75** of the inner cover **61** exist in front of the opening **53**, that is, in front of the toner container **39**.

[0072] The user then pulls out the toner container **39** forward through the opening **53**, as shown in FIG. **13**. At this time, the toner container **39** comes into contact with the protruding portions **75** of the inner cover **61**. Then, the protruding portions **75** are pushed by the toner container **39**, and the inner cover **61** is turned further downward. At this time, since the first protrusion **83** of the inner cover **61**

pushes up the lower stopper piece **105**, the lower stopper piece **105** is elastically turned upward.

[0073] The inner cover **61** is turned until the second protrusion **85** abuts the engagement portion **125** of the upper stopper piece **103** (see the arrow in FIG. **13**). At this time, an opening angle of the inner cover **61** is 120°. By turning the inner cover **61** in this manner, the toner container **39** can be pulled out through the opening **53**. The upper stopper piece **103** is an example of the second stopper which holds the inner cover **61** at an angle of larger than 90°.

[0074] In addition, when the inner cover **61** is turned in this manner, the upper corner portion of the lower inner cover **61** enters the corner between the lower portion **71***a* and the inclined portion **71***c* of the cover plate **71** of the inner cover **61**. As described above, the lower portion **71***a* of the cover plate **71** is an example of the recess in which the upper end portion of the lower inner cover **61** enters when the inner cover **61** is turned in the open posture. Thus, even when the upper inner cover **61** and the lower inner cover **61** are close to each other in the upper-and-lower direction so that the rotation range of the upper inner cover **61** and the rotation range of the lower inner cover **61** partially overlap, the upper inner cover **61** can be turned at an opening angle of 120°.

[0075] When the toner container **39** is completely pulled out, the lower stopper piece **105** elastically returns to bias the inner cover **61** upward. As a result, the inner cover **61** returns to the open posture (see FIG. **12**) in which the opening angle is 90°.

[0076] The user detaches the empty toner container **39** and attaches a new toner container **39** to the container housing part **41** through the opening **53**. Each toner container **39** is attached to the container housing part **41** in an attachment direction along the front-and-rear direction, with the end portion provided with the discharge port **39***b* facing forward (the rear side). Then, the attachment of the container is detected, and the various information is read from the memory chip and then transmitted to the control part (not shown). The control part stores the detection result of the toner sensor **55** and the various information read from the toner container **39** in correspondence with each other.

[0077] After the toner container **39** is attached, the inner cover **61** is turned upward. Then, the hook **79** of the inner cover **61** presses the hook piece **141***b* of the hook member **141** of the locking device **93**. As described above, since there is a play between the shaft **143***b* provided at the distal end of the movable member **143***a* and the opening **141***d* provided in the hook member **141**, the hook member **141** is turned in the release direction B, and after the inner cover **61** is turned in the close posture completely, the hook member **141** is turned in the engagement direction A by being biased by the torsion coil spring **145**, and the hook piece **141***b* is engaged with the hook **79** of the inner cover **61**. That is, the inner cover **61** is locked to the front side plate **51** by the locking device **93**.

[0078] When the inner cover **61** is turned in the close posture as described above, the toner container **39** is pushed in by the inner cover **61**. Thus, the toner container **39** can be surely attached to the container housing part **41**. When the toner container **39** is rotated at the time of supplying the toner from the toner container **39** to the developing device **35**, the toner container **39** may move in the front-and-rear direction. In this case, the movement of the toner container **39** in the forward direction is restricted by the inner cover **61**.

[0079] Thereafter, the user turns the outer cover **63** to the close posture. Then, the control part drives the rotation mechanism of the toner supply mechanism in response to the detection of the attachment of the toner container **39** and the turning of the outer cover **63** in the close posture. As a result, the toner container **39** rotates, and the contained toner is conveyed toward the discharge port **39***b* by the spiral protrusion **39***c*. The toner is discharged from the discharge port **39***b* to the pipe, and supplied to the corresponding developing device **35** through the pipe.

[0080] When the toner sensor **55** detects that the toner is supplied, the rotation drive of the toner container **39** is stopped. When the toner is not detected by the toner sensor **55** within a predetermined time, it is determined that the toner container is empty and the rotation drive of the toner container **39** is stopped. Therefore, when the outer cover **63** is opened, the solenoid **143** is energized again, and the inner cover **61** is opened.

[0081] When the toner container **39** is replaced, one empty toner container **39** may be replaced, and the two toner containers **39** may be replaced simultaneously after the two toner containers **39** become empty. When one toner container **39** is replaced, it is replaced as described above.

[0082] When the two toner containers **39** are replaced simultaneously, when the user turns the outer cover **63** in the open posture, the upper and lower inner covers **61** are automatically turned in the open posture simultaneously. At this time, both the upper and lower inner covers **61** are turned at the opening angle of 90°. When the toner container **39** is pulled out, as described above, the inner cover **61** is pushed by the toner container **39** and turned further downward, so that the toner container **39** can be pulled out. When the upper toner container **39** is pulled out, as described above, the inner cover **61** is turned to the opening angle of 120°, but after pulling out, the lower stopper piece **105** elastically returns and the inner cover **61** is turned to the opening angle of 90°. Therefore, when the lower toner container **39** is pulled out, the inner cover **61** does not exist in the space in front of the toner container **39**. Accordingly, the lower toner container **39** can be pulled out without interfering with the upper inner cover **61** turned in the open posture.

[0083] When the toner container **39** is pulled out, the R-surface shaped upper corner of the inner cover **61** slides with respect to the toner container **39**, so that the toner container **39** is smoothly detached without being caught on the inner cover **61**.

[0084] As is clear from the above description, according to the image forming apparatus **1** of the present disclosure, even when the inner covers **61** are arranged in the upper-and-lower direction and the left-and-right direction almost without any gap, and the rotation range of the upper inner cover **61** and the rotation range of the lower inner cover **61** overlap, the upper and lower toner containers **39** can be smoothly attached and detached without interfering with the inner cover **61** turned in the open posture.

[0085] Specifically, when the lock of the inner cover **61** by the locking device **93** is released, the inner cover **61** is automatically turned and held at the opening angle of 90°. When the toner container **39** is pulled out and the inner cover **61** is pushed down, the inner cover **61** is turned to the opening angle of 120°, but after the toner container **39** is pulled out, the inner cover **61** is automatically turned to the opening angle of 90°. Therefore, when the lower toner container **39** below the pulled out upper toner container **39** is pulled out, the toner container **39** is prevented from interfering with the upper inner cover **61**.

[0086] When the upper inner cover **61** is turned in the open posture, the upper corner of the lower inner cover **61** enters the lower portion **71***a* of the cover plate **71** of the inner cover **61**, so that the upper inner cover **61** can also be turned to the opening angle of 90° or larger.

[0087] Further, when the toner sensor **55** detects that the toner container **39** becomes empty, when the outer cover **63** is opened, the locking device **93** of the inner cover **61** for opening and closing the opening **53** corresponding to the toner container **39** releases the locking of the inner cover **61**. In a state where the inner cover **61** is turned in the close posture, the center of gravity of the inner cover **61** is positioned on the outer side of the front side plate **51** than the inner rotational shafts **65**. Accordingly, when the locking of the inner cover **61** by the locking device **93** is released, the inner cover **61** is automatically turned in the open posture, thereby improving the workability of attaching and detaching the toner container **39**.

[0088] The upper and lower stopper pieces **103** and **105** are cantilevered by the right and left side plates **101**, and are integrally formed as the stopper member **91**. Therefore, both pieces can be easily positioned and mounted in the housing space **139**. Further, since the upward turning of is restricted, after the inner cover **61** is turned until the second protrusion **85** is engaged with the upper stopper piece **103**, further turning of the inner cover **61** is restricted. Therefore, excessive turning of the inner cover **61** can be restricted.

[0089] The opening angle of the inner cover **61** is not limited to 90° and 120°. One opening angle may be less than 90° and the other opening angle may be larger than 90°. The inner cover **61** may be turned to two or more opening angles.

[0090] The container housing part **41** may be provided to house three or more container groups. Further, the openings in the upper and lower opening rows need not be displaced in the left-and-right direction.

[0091] While the present disclosure has been described for specific embodiments, the present disclosure is not limited to those embodiments. Without departing from the scope and intent of this disclosure.

1. An image forming apparatus comprising:

a plurality of container groups each including a plurality of toner containers containing of toner of different colors; and

a housing part to and from which the toner container is attached and detached, wherein

the housing part includes:

a side plate in which opening rows are formed for each container group in a plurality of levels in an upper-and-lower direction, the opening row including a plurality of openings through which the toner containers are attached and detached and which are arranged horizontally;

a cover which is supported by a rotational shaft provided at a lower end portion of each opening of the side plate, and is turned in an open posture for opening the opening and a close posture for closing the opening;

a locking device which locks the cover turned in the close posture to the side plate; and

a first stopper which holds the cover at an opening angle of 90° or less to the side plate and a second stopper which holds the cover at an opening angle of larger than 90° when the cover is turned to the open posture.

2. The image forming apparatus according to claim **1**, wherein

the first stopper and the second stopper are spaced apart in a circumferential direction of the rotational shaft, and

the first stopper is elastically deformed to be turned upward.

3. The image forming apparatus according to claim **2**, wherein

the cover has a first protrusion and a second protrusion spaced apart in the circumferential direction of the rotational shaft,

when a locking of the cover by the locking device is released, the cover is turned by its own weight until the first protrusion is engaged with the first stopper, and thereafter, when the cover is pushed downward, the first stopper is pushed by the first protrusion to be elastically deformed to be turned, and the cover is turned until the second protrusion is engaged with the second stopper.

4. The image forming apparatus according to claim **2**, wherein

when the cover is released after being pushed down, the cover is turned until stopper is elastically returned and the first protrusion is engaged with the first stopper.

5. The image forming apparatus according to claim **1**, wherein

the first stopper and the second stopper are integrally formed.

6. The image forming apparatus according to claim **3**, wherein

the cover is pushed down by the toner container pulled out through the opening.

7. The image forming apparatus according to claim **1**, wherein

the cover has a recess recessed in a thickness direction of the cover, and an upper end portion of the lower cover enters the recess when the upper cover is turned in the open posture.

8. The image forming apparatus according to claim **1**, wherein

a center of gravity of the cover is positioned on an outer side of the side plate than the rotational shaft when the cover is turned in the close posture.

9. The image forming apparatus according to claim **1**, comprising:

a sensor which detects that the toner container becomes empty, wherein

when the sensor detects that the toner container becomes empty, a locking of the cover by the locking device corresponding to the cover which opens and closes the opening corresponding to the toner container is released.

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